615HW5

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615 Strawberry data cleaning

```r
# Load the strawberry dataset and take a glimpse at the data
strawberry_data <- read_csv("strawberries25_v3.csv", col_names = TRUE)
```

```
## Rows: 12669 Columns: 21
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (15): Program, Period, Geo Level, State, State ANSI, Ag District, County...
## dbl  (2): Year, Ag District Code
## lgl  (4): Week Ending, Zip Code, Region, Watershed
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
glimpse(strawberry_data)
```

```
## Rows: 12,669
## Columns: 21
## $ Program             <chr> "CENSUS", "CENSUS", "CENSUS", "CENSUS", "CENSUS", "~
## $ Year                <dbl> 2022, 2022, 2022, 2022, 2022, 2022, 2022, 2022, 202~
## $ Period              <chr> "YEAR", "YEAR", "YEAR", "YEAR", "YEAR", "YEAR", "YE~
## $ `Week Ending`       <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ `Geo Level`         <chr> "COUNTY", "COUNTY", "COUNTY", "COUNTY", "COUNTY", "~
## $ State               <chr> "ALABAMA", "ALABAMA", "ALABAMA", "ALABAMA", "ALABAM~
## $ `State ANSI`        <chr> "01", "01", "01", "01", "01", "01", "01", "01", "01~
## $ `Ag District`       <chr> "BLACK BELT", "BLACK BELT", "BLACK BELT", "BLACK BE~
## $ `Ag District Code`  <dbl> 40, 40, 40, 40, 40, 40, 40, 40, 40, 40, 40, 40, 40,~
## $ County              <chr> "BULLOCK", "BULLOCK", "BULLOCK", "BULLOCK", "BULLOC~
## $ `County ANSI`       <chr> "011", "011", "011", "011", "011", "011", "101", "1~
## $ `Zip Code`          <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ Region              <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ watershed_code      <chr> "00000000", "00000000", "00000000", "00000000", "00~
## $ Watershed           <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ Commodity           <chr> "STRAWBERRIES", "STRAWBERRIES", "STRAWBERRIES", "ST~
## $ `Data Item`         <chr> "STRAWBERRIES - ACRES BEARING", "STRAWBERRIES - ACR~
## $ Domain              <chr> "TOTAL", "TOTAL", "TOTAL", "TOTAL", "TOTAL", "TOTAL~
## $ `Domain Category`   <chr> "NOT SPECIFIED", "NOT SPECIFIED", "NOT SPECIFIED", ~
## $ Value               <chr> "(D)", "3", "(D)", "1", "6", "5", "(D)", "(D)", "2"~
## $ `CV (%)`            <chr> "(D)", "15.7", "(D)", "(L)", "52.7", "47.6", "(D)",~
```

Explanation

Dataset Overview: The `strawberry_data` contains 12,699 rows and 21 columns. Columns include information like date, location, values, and coefficients of variation.

Remove Columns with Single Value in All Rows

```r
#|label: function def - drop 1-item columns

# Function to remove columns with only a single unique value
drop_one_value_col <- function(df) {
  drop <- NULL

  # Test each column for a single value
  for (i in 1:ncol(df)) {
    if (n_distinct(df[[i]]) == 1) {
      drop <- c(drop, i)
    }
  }

  # Report the result - names of columns dropped
  if (is.null(drop)) {
    return(df)
  } else {
    print("Columns dropped:")
    print(colnames(df)[drop])
    df <- df[, -drop]
  }
}

# Use the function to remove single-value columns
strawberry_data <- drop_one_value_col(strawberry_data)
```

```
## [1] "Columns dropped:"
## [1] "Week Ending"    "Zip Code"       "Region"         "watershed_code"
## [5] "Watershed"      "Commodity"
```

```r
glimpse(strawberry_data)
```

```
## Rows: 12,669
## Columns: 15
## $ Program             <chr> "CENSUS", "CENSUS", "CENSUS", "CENSUS", "CENSUS", "~
## $ Year                <dbl> 2022, 2022, 2022, 2022, 2022, 2022, 2022, 2022, 202~
## $ Period              <chr> "YEAR", "YEAR", "YEAR", "YEAR", "YEAR", "YEAR", "YE~
## $ `Geo Level`         <chr> "COUNTY", "COUNTY", "COUNTY", "COUNTY", "COUNTY", "~
## $ State               <chr> "ALABAMA", "ALABAMA", "ALABAMA", "ALABAMA", "ALABAM~
## $ `State ANSI`        <chr> "01", "01", "01", "01", "01", "01", "01", "01", "01~
## $ `Ag District`       <chr> "BLACK BELT", "BLACK BELT", "BLACK BELT", "BLACK BE~
## $ `Ag District Code`  <dbl> 40, 40, 40, 40, 40, 40, 40, 40, 40, 40, 40, 40, 40,~
## $ County              <chr> "BULLOCK", "BULLOCK", "BULLOCK", "BULLOCK", "BULLOC~
## $ `County ANSI`       <chr> "011", "011", "011", "011", "011", "011", "101", "1~
```

```
## $ ‘Data Item‘        <chr> "STRAWBERRIES - ACRES BEARING", "STRAWBERRIES - ACR~
## $ Domain             <chr> "TOTAL", "TOTAL", "TOTAL", "TOTAL", "TOTAL", "TOTAL~
## $ ‘Domain Category‘  <chr> "NOT SPECIFIED", "NOT SPECIFIED", "NOT SPECIFIED", ~
## $ Value              <chr> "(D)", "3", "(D)", "1", "6", "5", "(D)", "(D)", "2"~
## $ ‘CV (%)‘           <chr> "(D)", "15.7", "(D)", "(L)", "52.7", "47.6", "(D)",~
```

Separate Composite Columns

```
#|label: split Data Item

# Split 'Data Item' column into two parts using delimiter '-'
strawberry_data <- strawberry_data |>
  separate_wider_delim(  cols = `Data Item`,
                         delim = "-",
                         names = c("column1", "column2"),
                         too_many = "merge",
                         too_few = "align_start")

# Further split 'column1' into 'Fruit' and 'Category' using delimiter ','
strawberry_data <- strawberry_data |>
  separate_wider_delim(  cols = `column1`,
                         delim = ",",
                         names = c("Fruit", "Category"),
                         too_many = "merge",
                         too_few = "align_start")

# Trim white spaces from the split columns
strawberry_data$Fruit <- str_trim(strawberry_data$Fruit, side = "both")
strawberry_data$Category <- str_trim(strawberry_data$Category, side = "both")
strawberry_data$column2 <- str_trim(strawberry_data$column2, side = "both")

# Remove columns with single-value again after modifications
strawberry_data <- drop_one_value_col(strawberry_data)
```

```
## [1] "Columns dropped:"
## [1] "Fruit"
```

```
unique(strawberry_data$Category)
```

```
##  [1] NA                        "ORGANIC"                 "ORGANIC, FRESH MARKET"
##  [4] "ORGANIC, PROCESSING"     "FRESH MARKET"            "PROCESSING"
##  [7] "FRESH MARKET, UTILIZED" "NOT SOLD"                "PROCESSING, UTILIZED"
## [10] "UTILIZED"                "BEARING"
```

Clean Data in the ‘Category’ Column

```
#|label: Clean data in the Category

# Extract specific attributes from 'Category' and assign them to new columns
```

```r
strawberry_data <- strawberry_data %>%
  mutate(Marketing_channels = ifelse(str_detect(Category, "FRESH MARKET|PROCESSING"),
                                     str_extract(Category, "FRESH MARKET|PROCESSING"), NA)) %>%
  mutate(Utilizations = ifelse(str_detect(Category, "UTILIZED"),
                               str_extract(Category, "UTILIZED"), NA)) %>%
  mutate(Method = ifelse(str_detect(Category, "ORGANIC"),
                         str_extract(Category, "ORGANIC"), NA)) %>%
  mutate(Class = ifelse(str_detect(Category, "BEARING"),
                        str_extract(Category, "BEARING"), NA)) %>%
  mutate(Measurement = ifelse(str_detect(Category, "NOT SOLD"),
                              str_extract(Category, "NOT SOLD"), NA)) %>%
  select(1:match("County ANSI", names(.)), Class, Method, Marketing_channels, Utilizations, Measurement
  select(-Category)
```

Clean Data in 'column2'

```r
#|label: Clean column2

# Split 'column2' into 'Measurement1' and 'Metric1'
strawberry_data <- strawberry_data %>%
  mutate(
    Measurement1 = ifelse(str_detect(column2, ","),
                          str_split_fixed(column2, ",", 2)[, 1],
                          column2),
    Metric1 = ifelse(str_detect(column2, ","),
                     str_split_fixed(column2, ",", 2)[, 2],
                     NA)
  ) %>%
  select(1:match("Measurement", names(.)), Measurement1, Metric1, everything()) %>%
  select(-column2)
```

Classify Strings in Their Positions

```r
#|label: classify strings

# Extract specific strings from 'Metric1' to 'Metric' and 'Remark'
strawberry_data <- strawberry_data %>%
  mutate(Metric = str_extract(Metric1, "(?<=,|^)[^,]*MEASURED IN[^,]*(?=,|$)")) %>%
  mutate(Remark = str_remove_all(Metric1, "(?<=,|^)[^,]*MEASURED IN[^,]*(,)?")) %>%
  mutate(Remark = str_trim(str_replace_all(Remark, "^,|,$|,,", ""))) %>%
  select(1:match("Measurement1", names(.)), Metric, Remark, everything())  %>%
  select(-Metric1)

# Create 'Category' column by combining 'Measurement' and 'Measurement1'
strawberry_data <- strawberry_data %>%
  mutate(Category = ifelse(is.na(Measurement),
                           Measurement1,
                           paste(Measurement, Measurement1, sep = " "))) %>%
  select(1:match("Utilizations", names(.)), Category, everything()) %>%
```

```r
  select(-Measurement, -Measurement1)

# Trim whitespace
strawberry_data$Metric <- str_trim(strawberry_data$Metric, side = "both")
strawberry_data$Category <- str_trim(strawberry_data$Category, side = "both")

# Replace empty strings in 'Remark' with NA
strawberry_data <- strawberry_data %>% mutate(across(Remark, ~ na_if(., "")))
glimpse(strawberry_data)
```

```
## Rows: 12,669
## Columns: 21
## $ Program             <chr> "CENSUS", "CENSUS", "CENSUS", "CENSUS", "CENSUS", "~
## $ Year                <dbl> 2022, 2022, 2022, 2022, 2022, 2022, 2022, 2022, 202~
## $ Period              <chr> "YEAR", "YEAR", "YEAR", "YEAR", "YEAR", "YEAR", "YE~
## $ `Geo Level`         <chr> "COUNTY", "COUNTY", "COUNTY", "COUNTY", "COUNTY", "~
## $ State               <chr> "ALABAMA", "ALABAMA", "ALABAMA", "ALABAMA", "ALABAM~
## $ `State ANSI`        <chr> "01", "01", "01", "01", "01", "01", "01", "01", "01~
## $ `Ag District`       <chr> "BLACK BELT", "BLACK BELT", "BLACK BELT", "BLACK BE~
## $ `Ag District Code` <dbl> 40, 40, 40, 40, 40, 40, 40, 40, 40, 40, 40, 40, 40,~
## $ County              <chr> "BULLOCK", "BULLOCK", "BULLOCK", "BULLOCK", "BULLOC~
## $ `County ANSI`       <chr> "011", "011", "011", "011", "011", "011", "101", "1~
## $ Class               <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ Method              <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ Marketing_channels <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ Utilizations        <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ Category            <chr> "ACRES BEARING", "ACRES GROWN", "ACRES NON-BEARING"~
## $ Metric              <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ Remark              <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ Domain              <chr> "TOTAL", "TOTAL", "TOTAL", "TOTAL", "TOTAL", "TOTAL~
## $ `Domain Category`  <chr> "NOT SPECIFIED", "NOT SPECIFIED", "NOT SPECIFIED", ~
## $ Value               <chr> "(D)", "3", "(D)", "1", "6", "5", "(D)", "(D)", "2"~
## $ `CV (%)`            <chr> "(D)", "15.7", "(D)", "(L)", "52.7", "47.6", "(D)",~
```

Separate Domain and Domain Category

```r
#|label: separate domain category

# Clean up 'Domain Category' by removing redundant text and characters
strawberry_data <- strawberry_data %>%
  mutate(`Domain Category` = ifelse(str_detect(`Domain Category`, ":"),
                              str_split_fixed(`Domain Category`, ":", 2)[, 2],
                              `Domain Category`)) %>%
  mutate(`Domain Category` = str_replace_all(`Domain Category`, "[\\(\\)]", ""))

strawberry_data$`Domain Category` <- str_trim(strawberry_data$`Domain Category`, side = "both")

# Separate 'Domain Category' into specific chemical and numbers
strawberry_data <- strawberry_data %>%
  mutate(Chemical_Number = ifelse(str_detect(`Domain Category`, "="),
                                  str_split_fixed(`Domain Category`, "=", 2)[, 2],
```

```r
                                  NA)) %>%
  mutate(`Domain Category` = ifelse(str_detect(`Domain Category`, "="),
                             str_split_fixed(`Domain Category`, "=", 2)[, 1],
                             `Domain Category`)) %>%
  select(1:match("Domain Category", names(.)), Chemical_Number, everything())

# Trim whitespace
strawberry_data$`Domain Category` <- str_trim(strawberry_data$`Domain Category`, side = "both")
strawberry_data$Chemical_Number <- str_trim(strawberry_data$Chemical_Number, side = "both")

# Remove 'CHEMICAL' from the 'Domain' column
strawberry_data <- strawberry_data %>%
  mutate(Domain = str_replace(Domain, "CHEMICAL, ", ""))
glimpse(strawberry_data)
```

```
## Rows: 12,669
## Columns: 22
## $ Program             <chr> "CENSUS", "CENSUS", "CENSUS", "CENSUS", "CENSUS", "~
## $ Year                <dbl> 2022, 2022, 2022, 2022, 2022, 2022, 2022, 2022, 202~
## $ Period              <chr> "YEAR", "YEAR", "YEAR", "YEAR", "YEAR", "YEAR", "YE~
## $ `Geo Level`         <chr> "COUNTY", "COUNTY", "COUNTY", "COUNTY", "COUNTY", "~
## $ State               <chr> "ALABAMA", "ALABAMA", "ALABAMA", "ALABAMA", "ALABAM~
## $ `State ANSI`        <chr> "01", "01", "01", "01", "01", "01", "01", "01", "01~
## $ `Ag District`       <chr> "BLACK BELT", "BLACK BELT", "BLACK BELT", "BLACK BE~
## $ `Ag District Code`  <dbl> 40, 40, 40, 40, 40, 40, 40, 40, 40, 40, 40, 40, 40,~
## $ County              <chr> "BULLOCK", "BULLOCK", "BULLOCK", "BULLOCK", "BULLOC~
## $ `County ANSI`       <chr> "011", "011", "011", "011", "011", "011", "101", "1~
## $ Class               <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ Method              <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ Marketing_channels  <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ Utilizations        <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ Category            <chr> "ACRES BEARING", "ACRES GROWN", "ACRES NON-BEARING"~
## $ Metric              <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ Remark              <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ Domain              <chr> "TOTAL", "TOTAL", "TOTAL", "TOTAL", "TOTAL", "TOTAL~
## $ `Domain Category`   <chr> "NOT SPECIFIED", "NOT SPECIFIED", "NOT SPECIFIED", ~
## $ Chemical_Number     <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ Value               <chr> "(D)", "3", "(D)", "1", "6", "5", "(D)", "(D)", "2"~
## $ `CV (%)`            <chr> "(D)", "15.7", "(D)", "(L)", "52.7", "47.6", "(D)",~
```

Transfer Data Types to Correct Types

```r
#|label: transfer data types

# Replace specific strings in 'Value' and 'CV (%)' columns and convert to numeric
strawberry_data <- strawberry_data %>%
  mutate(Value = ifelse(Value %in% c("(D)", "(NA)"), NA, Value)) %>%
  mutate(Value = ifelse(Value == "(Z)", "0.0005", Value)) %>%
  mutate(Value = str_replace_all(Value,",","")) %>%
  mutate(Value = as.numeric(Value))
```

```r
strawberry_data <- strawberry_data %>%
  mutate(`CV (%)` = ifelse(`CV (%)` %in% c("(D)", "(NA)"), NA, `CV (%)`)) %>%
  mutate(`CV (%)` = ifelse(`CV (%)` == "(L)", "0.05", `CV (%)`)) %>%
  mutate(`CV (%)` = ifelse(`CV (%)` == "(H)", "99.95", `CV (%)`)) %>%
  mutate(`CV (%)` = str_replace_all(`CV (%)`,",","")) %>%
  mutate(`CV (%)` = as.numeric(`CV (%)`))
glimpse(strawberry_data)
```

```
## Rows: 12,669
## Columns: 22
## $ Program            <chr> "CENSUS", "CENSUS", "CENSUS", "CENSUS", "CENSUS", "~
## $ Year               <dbl> 2022, 2022, 2022, 2022, 2022, 2022, 2022, 2022, 202~
## $ Period             <chr> "YEAR", "YEAR", "YEAR", "YEAR", "YEAR", "YEAR", "YE~
## $ `Geo Level`        <chr> "COUNTY", "COUNTY", "COUNTY", "COUNTY", "COUNTY", "~
## $ State              <chr> "ALABAMA", "ALABAMA", "ALABAMA", "ALABAMA", "ALABAM~
## $ `State ANSI`       <chr> "01", "01", "01", "01", "01", "01", "01", "01", "01~
## $ `Ag District`      <chr> "BLACK BELT", "BLACK BELT", "BLACK BELT", "BLACK BE~
## $ `Ag District Code` <dbl> 40, 40, 40, 40, 40, 40, 40, 40, 40, 40, 40, 40, 40,~
## $ County             <chr> "BULLOCK", "BULLOCK", "BULLOCK", "BULLOCK", "BULLOC~
## $ `County ANSI`      <chr> "011", "011", "011", "011", "011", "011", "101", "1~
## $ Class              <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ Method             <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ Marketing_channels <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ Utilizations       <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ Category           <chr> "ACRES BEARING", "ACRES GROWN", "ACRES NON-BEARING"~
## $ Metric             <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ Remark             <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ Domain             <chr> "TOTAL", "TOTAL", "TOTAL", "TOTAL", "TOTAL", "TOTAL~
## $ `Domain Category`  <chr> "NOT SPECIFIED", "NOT SPECIFIED", "NOT SPECIFIED", ~
## $ Chemical_Number    <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ Value              <dbl> NA, 3, NA, 1, 6, 5, NA, NA, 2, 2, NA, NA, 2, 2, 1, ~
## $ `CV (%)`           <dbl> NA, 15.70, NA, 0.05, 52.70, 47.60, NA, NA, 55.70, 5~
```

Separate the Data into Categories

```r
#|label: separate data categories

# Split the table into 'census' and 'survey'
strawberry_census <- strawberry_data %>% filter(Program == "CENSUS")
strawberry_survey <- strawberry_data %>% filter(Program == "SURVEY")
strawberry_census <- drop_one_value_col(strawberry_census)
```

```
## [1] "Columns dropped:"
## [1] "Program"         "Period"          "Class"           "Utilizations"
## [5] "Remark"          "Chemical_Number"
```

```r
strawberry_survey <- drop_one_value_col(strawberry_survey)
```

```
## [1] "Columns dropped:"
## [1] "Program"          "Ag District"      "Ag District Code" "County"
## [5] "County ANSI"      "Method"           "CV (%)"
```

```r
# Split the 'census' table into 'organic' and 'non-organic'
strawberry_organic <- strawberry_census %>% filter(Domain == "ORGANIC STATUS")
strawberry_organic <- drop_one_value_col(strawberry_organic)
```

```
## [1] "Columns dropped:"
## [1] "Ag District"      "Ag District Code" "County"           "County ANSI"
## [5] "Method"           "Domain"           "Domain Category"
```

```r
strawberry_non_organic <- strawberry_census %>% filter(!Domain == "ORGANIC STATUS")
strawberry_non_organic <- drop_one_value_col(strawberry_non_organic)
```

```
## [1] "Columns dropped:"
## [1] "Year"               "Method"             "Marketing_channels"
## [4] "Metric"
```

```r
# Split the 'survey' table into 'chemical' and 'non-chemical'
strawberry_chemical <- strawberry_survey %>% filter(!Domain == "TOTAL")
strawberry_chemical <- drop_one_value_col(strawberry_chemical)
```

```
## [1] "Columns dropped:"
## [1] "Period"             "Geo Level"          "Marketing_channels"
## [4] "Utilizations"
```

```r
strawberry_non_chemical <- strawberry_survey %>% filter(Domain == "TOTAL")
strawberry_non_chemical <- drop_one_value_col(strawberry_non_chemical)
```

```
## [1] "Columns dropped:"
## [1] "Class"           "Domain"          "Domain Category" "Chemical_Number"
```

Estimate the NA in Value and CV(%)

```r
#|label: estimate missing values

# Function to estimate missing values in a given dataset using linear regression
estimate_na_values <- function(df, value_model_formula, cv_model_formula = NULL) {
  df <- df %>% mutate(original_order = row_number())
  # Value estimation
  with_value <- df %>% filter(!is.na(Value))
  missing_value <- df %>% filter(is.na(Value))
  value_model <- lm(value_model_formula, data = with_value)
  missing_value <- missing_value %>%
    mutate(Value = round(exp(predict(value_model, newdata = missing_value)), 1))
  df_filled <- bind_rows(with_value, missing_value) %>% arrange(original_order)
  # CV estimation
  if (!is.null(cv_model_formula)) {
    with_cv <- df_filled %>% filter(!is.na(`CV (%)`))
    missing_cv <- df_filled %>% filter(is.na(`CV (%)`))
    cv_model <- lm(cv_model_formula, data = with_cv)
    missing_cv <- missing_cv %>%
```

```r
    mutate(`CV (%)` = round(predict(cv_model, newdata = missing_cv), 2))
    df_filled <- bind_rows(with_cv, missing_cv) %>% arrange(original_order)
  }
  return(df_filled %>% select(-original_order) %>% mutate(Value = round(Value, 1)))
}

# Estimate missing values for 'strawberry_organic'
strawberry_organic <- estimate_na_values(strawberry_organic, log(Value) ~ factor(Year) + State + Catego

# Estimate missing values for 'strawberry_non_organic'
strawberry_non_organic <- estimate_na_values(strawberry_non_organic, Value ~ State + Category + Domain

# Estimate missing values for 'strawberry_chemical'
strawberry_chemical <- estimate_na_values(strawberry_chemical, log(Value) ~ factor(Year) + State + Cate

# Estimate missing values for 'strawberry_non_chemical'
strawberry_non_chemical <- estimate_na_values(strawberry_non_chemical, log(Value + 1) ~ factor(Year) +
```

Write Cleaned Data to CSV Files

```r
#|label: write csv

# Write cleaned datasets to new CSV files
write.csv(strawberry_organic, "strawberry_organic.csv", row.names = FALSE)
write.csv(strawberry_non_organic, "strawberry_non_organic.csv", row.names = FALSE)
write.csv(strawberry_chemical, "strawberry_chemical.csv", row.names = FALSE)
write.csv(strawberry_non_chemical, "strawberry_non_chemical.csv", row.names = FALSE)
```

Explanation of Changes

- Consolidated repetitive data cleaning steps to use reusable functions where possible.
- Streamlined data cleaning, trimming, and column separation using tidyverse functions.
- Implemented a generalized function `estimate_na_values()` for estimating missing values using regression.
- Split the dataset into four separate tables: organic, non-organic, chemical, and non-chemical data, then wrote each to a separate CSV file.